Financial Tweet Sentiment Classification

Text Mining Course Project

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Group 27

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1. Introduction

Classifying financial tweets as Bearish (0), Bullish (1), or Neutral (2) is a genuinely hard NLP problem. The signal is subtle, meaning that, the difference between "beats EPS" and "misses EPS" is a single word, yet it defines opposite market sentiments. The vocabulary is domain-specific, containing ticker symbols, financial jargon, and Twitter-specific noise that standard NLP tools were not designed for. The dataset is also heavily imbalanced given that the class *Neutral* dominates at 64.7% meaning, a naive majority-class classifier achieves deceptively high accuracy while completely failing on the classes that matter most to investors.

This report documents a full NLP pipeline developed to address this task, covering exploratory data analysis, preprocessing, feature engineering across three paradigms (TF-IDF, Word2Vec/GloVe, and Transformer Encoders), and classification using Logistic Regression, Random Forest, XGBoost, MLP, and fine-tuned transformer models (FinBERT, RoBERTa-base, Twitter-RoBERTa, and GPT-2).

2. Data Exploration

2.1. Dataset Overview

The training corpus consists of **9,543 tweets** with no missing values and no duplicate entries. Each tweet is labelled as Bearish (0), Bullish (1), or Neutral (2). The test set contains 2388 unlabelled tweets for which predictions must be submitted.

2.1.1. Class Imbalance

As shown in Figure the dataset is heavily imbalanced. The Neutral class dominates with **64.7%** of all examples (6,178 tweets), while Bullish accounts for **20.2%** (1,923) and Bearish for **15.1%** (1,442). The minority classes together represent just over one third of the data.

This imbalance has direct methodological consequences. A naïve classifier that always predicts Neutral would achieve approximately 65% accuracy which in itself is a misleadingly high score that reflects class frequency rather than genuine discriminative ability. For this reason, **macro-averaged F1-score** is adopted as the primary evaluation metric throughout this project, as it weights each class equally regardless of support. All train/validation splits were performed with **stratification** to preserve the original class proportions across subsets.

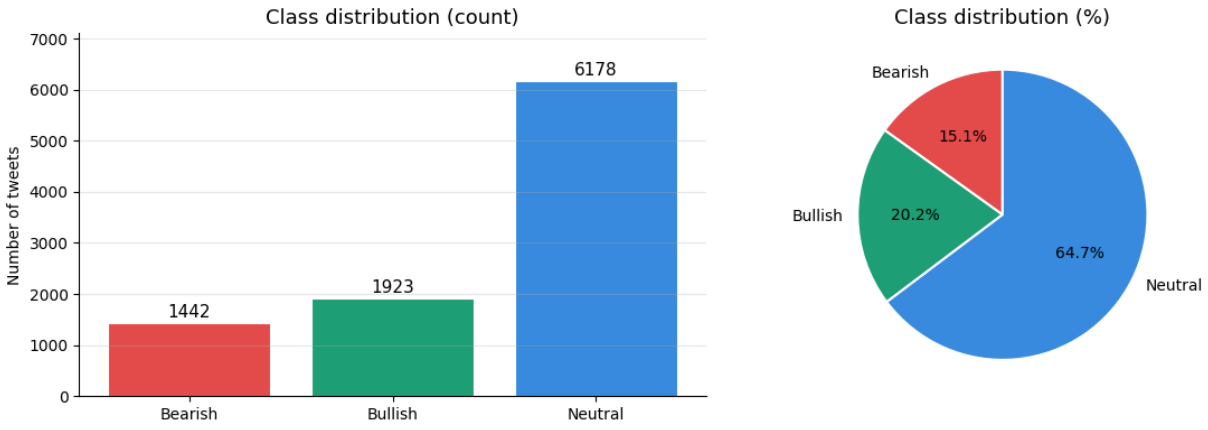


Figure 1

2.1.2. Tweet Length

Mean tweet length is nearly identical across all three classes (~12 words), confirming that length alone carries no discriminative signal. Vocabulary analysis revealed strongly class-specific term patterns:

- **Bullish-specific:** beats (164 occurrences), raised, higher, buy — positive earnings and price-action language
- **Bearish-specific:** misses (85 occurrences), cut, coronavirus — negative earnings and risk language
- **Neutral-specific:** declares, reports, dividend, earnings — announcement-style newswire register suggesting Neutral tweets are largely factual disclosures rather than absent sentiment

General financial terms such as stock and market appear uniformly across all classes and carry negligible discriminative value.

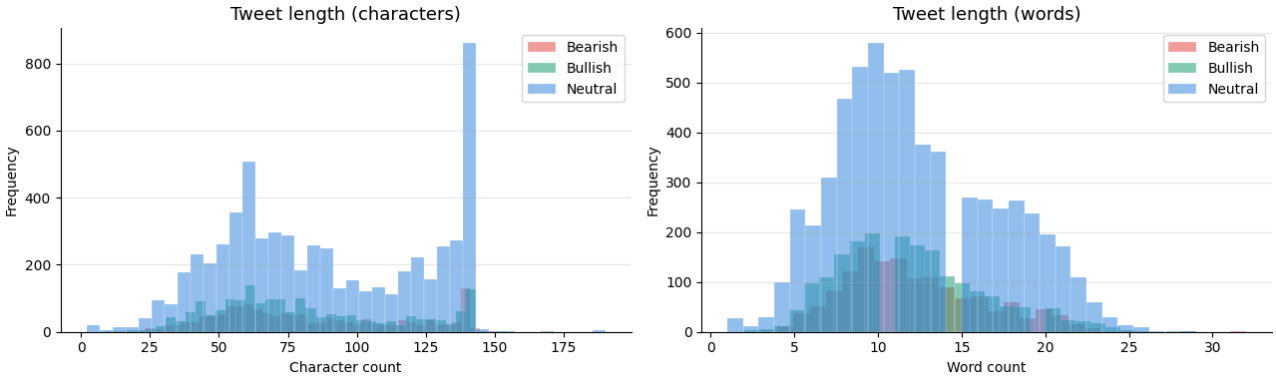


Figure 2

2.1.3. Vocabulary Analysis

Figure 3 and Figure illustrate the most frequent tokens per class after basic tokenisation.
Vocabulary is strongly class-discriminative. Despite sharing a common financial domain, Bullish and Bearish tweets exhibit near-opposite lexical profiles around the same underlying events. The term **"beats"** (164 occurrences, exclusive to Bullish tweets) is among the strongest positive indicators, capturing earnings surprises to the upside. Conversely, **"misses"** (85 occurrences, Bearish only) marks the downside equivalent. This antonymic pairing around earnings events is a recurring structural pattern in the corpus.
Analyst action verbs further differentiate the classes: Bullish tweets favour terms such as *raised*, *higher*, and *buy*, signalling upward revisions and recommendations, while Bearish tweets concentrate around *cut* and *coronavirus*, reflecting downgrades and macroeconomic distress signals.
Neutral tweets present a qualitatively distinct register. High-frequency terms such as *declares*, *reports*, *dividend*, and *earnings* point to a newswire or press-release style of writing which is factual, announcement-oriented, and deliberately lacking of evaluative language. This is a meaningful finding: **Neutral does not represent the absence of Bullish or Bearish content, but rather a separate communicative genre**. Models that treat it merely as a residual category are likely to underperform on this class.

Generic financial terms — *stock*, *market*, *share* — appear with high and approximately uniform frequency across all three classes and therefore carry little discriminative information.

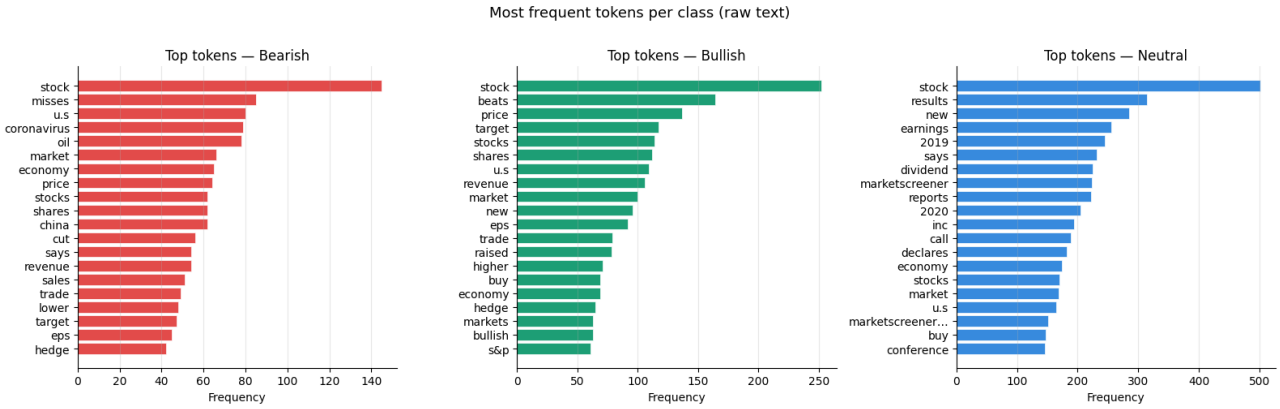


Figure 3



Figure 4

2.1.4. Special Token Analysis

Figure 5 reports the per-class prevalence of cashtags ($TICKER), URLs, and user mentions (@user).



Figure 5

Bullish tweets contain cashtags at **nearly twice the rate** of the other classes. This suggests that positive sentiment in financial Twitter is more frequently directed at a named, specific asset which is consistent with the behaviour of traders expressing conviction in individual

stocks. Cashtags thus represent a potentially informative feature and were **preserved during preprocessing**.
This finding, that positive sentiment is more frequently directed at specific named assets, directly influenced the preprocessing decision to **preserve cashtags** rather than remove them, unlike other Twitter-specific tokens. URLs appear uniformly across classes (39–50%) with no sentiment signal and were removed. User mentions are negligible (mean ≤ 0.05 per tweet).

2.1.5. Language and Emoji Analysis

Language detection was performed using the langdetect library. Approximately **7.2%** of tweets were flagged as non-English. However, manual inspection of a random sample of 30 such tweets revealed that the overwhelming majority were **false positives**: financial shorthand, stock ticker symbols, acronyms, and numerically dense text contain too little natural language for the detector to operate reliably. Only **2 genuinely non-English tweets** (in Italian) were identified in the sample.
Given the negligible true rate of non-English content, no tweets were excluded on the basis of language to avoid discarding valid English-language data.
Emoji presence was found to be negligible across all three classes, with a rate of at most **0.76%** per class. Their removal during preprocessing introduces no meaningful loss of information.

3. Data Preprocessing

Two preprocessing pipelines were developed: a full pipeline for traditional ML and word embedding methods, and a light pipeline for transformer-based models. This distinction is motivated by a fundamental difference in how these model families consume text: traditional approaches require normalised, reduced token vocabularies, whereas transformers such as FinBERT and RoBERTa are pre-trained on raw text and rely on their internal tokeniser and subword vocabulary — aggressive cleaning degrades rather than improves their input.

3.1. Corpus Split

Prior to any preprocessing, the training corpus was partitioned into a training set (80%, 7,634 examples) and a validation set (20%, 1,909 examples) using a stratified random split. Stratification ensures that the original class proportions, Bearish 15.1%, Bullish 20.2%, Neutral 64.7%, are preserved exactly across both subsets. This is particularly important given the severe class imbalance identified in exploration: an unstratified split could, by chance, under-represent the minority Bearish class in validation and produce an optimistically biased evaluation.
All preprocessing steps were fitted exclusively on the training set and applied identically to validation and test sets, preventing any form of data leakage.

3.2. Full Preprocessing Pipeline

The full pipeline applies four techniques in sequence and is used to prepare input for BoW, TF-IDF, Word2Vec, and traditional ML classifiers.

Technique 1 — Regex Cleaning

A set of regular expression rules removes noise whose absence of discriminative value was confirmed during exploration. The table below summarises each rule and its justification.

Pattern	Action	Justification
URLs (`http`, `www`)	Removed	Uniformly distributed across classes (39–50%); no sentiment signal
`@mentions`	Removed	Mean ≤ 0.046 occurrences per tweet across all classes
RT prefix	Removed	Structural Twitter retweet artifact; not authored content
`&`	Removed	HTML entity artifact from Twitter's API encoding
Hashtags (`#word`)	Removed	No meaningful class-level difference in presence (6.2–11.0%)
Non-ASCII / emojis	Removed	Emoji rate ≤ 0.76% across all classes; negligible information
Cashtags (`$AAPL`)	Preserved	Bullish tweets exhibit 2× the cashtag rate of other classes (25.0% vs ~13%); confirmed discriminative signal

Cashtags receive special handling: before any further processing, they are temporarily replaced with a TICKER_XXX prefix to protect them from being altered or removed by downstream steps, then restored to their original $xxx form at the end of the pipeline.

Technique 2 — Tokenization

NLTK's TweetTokenizer was selected over generic whitespace splitting. Financial Twitter text contains contractions, elongated words, and punctuation attached directly to tokens (e.g., "revenue.", "stock,"). TweetTokenizer handles these correctly and was designed for this register of text. The reduce_len=True parameter collapses exaggerated character repetitions (e.g., "soooo" → "soo"), which, while infrequent, avoids spurious vocabulary entries.

Technique 3 — Stop-word Removal

Standard NLTK English stop words were removed, as they carry no sentiment signal. The stop-word list was extended with four custom tokens identified during exploration:

- marketscreener — appeared 224 times predominantly in Neutral tweets as a data source attribution artifact, not authored language
- rt, amp, via — residual Twitter and HTML encoding artifacts that survived the regex step in some edge cases

Cashtag tokens are explicitly exempted from stop-word removal, regardless of whether they match any stop-word pattern.

Technique 4 — Lemmatization

Lemmatisation reduces inflected word forms to their canonical base, which directly benefits the most discriminative tokens identified in exploration: *beats* → *beat*, *misses* → *miss*, *raised* → *raise*. This consolidates counts across tense and number variants without destroying the recognisability of the root form.

A POS-aware implementation was used: each token is first tagged with its part-of-speech using NLTK's averaged_perceptron_tagger, and the appropriate WordNet POS category (verb, adjective, adverb, or noun) is passed to WordNetLemmatizer. Default noun-only lemmatisation would fail to reduce verb forms correctly — a critical limitation given that verbs (*beat*, *raise*, *cut*) are among the strongest discriminative signals in the corpus. Lemmatisation was preferred over stemming because stemming produces truncated, non-

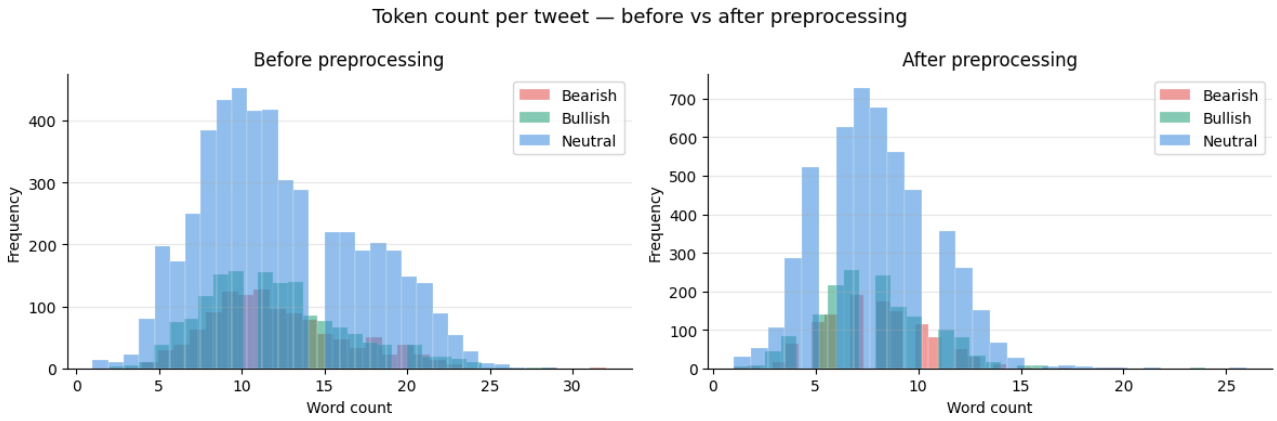
standard forms (e.g., *earns* → *earn* is fine, but *earnings* → *earn* via stemming loses the noun sense), which are also likely to fall outside the vocabulary of pre-trained embeddings used in later stages.

3.3. Light Preprocessing Pipeline (Transformer Input)

For transformer-based models (FinBERT, RoBERTa), only minimal cleaning is applied: URL removal, mention removal, RT prefix stripping, HTML entity decoding, hashtag removal, and non-ASCII/emoji removal. Crucially, **no lowercasing, no stop-word removal, and no lemmatisation** are performed. Transformer models are pre-trained on naturally occurring text; their subword tokenisers and contextual attention mechanisms are designed to operate on grammatically intact sentences. Removing stop words or normalising inflection strips the positional and syntactic cues that attention relies upon.

3.4. Effect on Token Count

Figure 6 shows the distribution of tweet length before and after applying the full pipeline. Mean token count decreases from **12.2 to 7.9 words per tweet**, a reduction of



approximately 35%, reflecting the removal of URLs, mentions, stop words, and punctuation. The distribution shape and the near-identical length across all three classes are preserved after preprocessing, confirming that the pipeline does not introduce any class-asymmetric distortion.

3.5. Handling of Empty Tweets

A small number of tweets were reduced to empty strings after preprocessing. These were tweets consisting almost entirely of URLs, mentions, or non-ASCII content with no remaining textual body. Empty entries were removed from the training and validation sets. For the test set, where removal is not permissible (all 299 predictions must be submitted), empty tweets were replaced with the placeholder token unk to allow the pipeline to produce a prediction without crashing.

4. Feature Engineering

Three families of feature engineering methods were implemented, each representing a distinct level of semantic expressiveness: sparse count-based representations (TF-IDF), dense corpus-trained embeddings (Word2Vec), and contextual transformer encoder embeddings (FinBERT, RoBERTa). Two variations were produced within each family, yielding six feature sets in total, which were subsequently used as inputs to the classification experiment.

4.1 Bag-of-Words / TF-IDF

TF-IDF (Term Frequency–Inverse Document Frequency) represents each tweet as a sparse vector over a fixed vocabulary, where each dimension reflects how distinctive a term is for a given document relative to the corpus. It is a strong baseline for short-text classification due to its simplicity and interpretability, and it operates directly on the preprocessed text produced by the full pipeline.

Two n-gram configurations were implemented:

Variation 1 — Unigrams (ngram_range=(1,1), max_features=20,000): Each feature corresponds to a single token. This is the canonical BoW representation and captures individual term salience. Given the discriminative vocabulary identified in exploration (*beats*, *misses*, *raised*, *cut*), individual tokens are expected to carry strong signal.

Variation 2 — Unigrams + Bigrams (ngram_range=(1,2), max_features=30,000): Extends the vocabulary with two-token sequences, capturing short phrases such as *price target*, *earnings beat*, or *stock price*. The feature space is wider (30,000 features) to accommodate the larger bigram vocabulary without crowding out unigrams.

Both variations share two common design choices motivated by the properties of the corpus. min_df=2 discards terms appearing in fewer than two documents, removing hapax legomena that would add noise without generalisation value. sublinear_tf=True applies a logarithmic transformation to raw term frequencies (tf → 1 + log(tf)), dampening the effect of terms that repeat many times within a single tweet — a common occurrence in short, repetitive financial text.

Both vectorisers were fitted exclusively on the training set and applied without refitting to validation and test sets.

4.2 Word2Vec (Dense Word Embeddings)

Word2Vec maps each token to a dense continuous vector in a learned semantic space, capturing distributional similarity between words. Unlike TF-IDF, it encodes the relationships between terms rather than their mere presence. Tweet-level representations are obtained by **mean pooling**, averaging the vectors of all in-vocabulary tokens in a tweet. Tweets with no in-vocabulary tokens are represented as zero vectors.

Two variations were implemented:

Variation 1 — Corpus-trained Word2Vec: A Word2Vec model was trained from scratch on the training corpus using the skip-gram architecture with a vector size of 100 dimensions, a context window of 5, a minimum count of 2, and 10 training epochs. Training on the in-domain corpus allows the model to learn representations specific to financial Twitter language, including the distributional behaviour of domain-specific terms and the protected cashtag tokens introduced in preprocessing.

Variation 2 — GloVe Twitter-100 (pre-trained): The glove-twitter-100 model, trained on 2 billion tweets with 100-dimensional vectors, was loaded via the Gensim downloader. This model provides broad coverage of Twitter-specific language and informal register and serves as an out-of-domain but transfer-rich baseline. Because the preprocessing pipeline lowercases all tokens, compatibility with GloVe's lowercase Twitter vocabulary is preserved.

The two variations enable a direct comparison between domain-specific distributional semantics (corpus-trained) and general Twitter-register semantics (GloVe), holding the pooling strategy and dimensionality constant.

4.3 Transformer Encoders

Transformer encoders produce **contextual embeddings**: the representation of each token is conditioned on the full surrounding sequence, allowing the same word to have different representations depending on context. This is a fundamental advantage over Word2Vec, where each token has a single static vector regardless of usage. Tweet-level representations were extracted using the **[CLS] token embedding** from the final hidden layer which is the standard approach for sentence-level tasks with BERT-family models. All

embeddings were extracted in inference mode (torch.no_grad()), processing the corpus in batches of 64 with a maximum sequence length of 128 tokens. The **light preprocessing pipeline** was used as input, preserving natural language structure for the transformer's subword tokeniser.

Variation 1 — FinBERT (ProsusAI/finbert): FinBERT is a BERT-base model fine-tuned on financial communications corpora including financial news, earnings call transcripts, and analyst reports. Its domain-specific pre-training makes it the most directly applicable encoder for this task: its vocabulary and attention patterns reflect financial language rather than general-purpose English. FinBERT produces 768-dimensional [CLS] embeddings.

Variation 2 — RoBERTa-base (roberta-base): RoBERTa is an optimised BERT pre-training procedure trained on a substantially larger general-domain corpus with dynamic masking. It provides a strong general-purpose contextual encoder baseline that is domain-agnostic but architecturally more robust than the original BERT. Comparing RoBERTa against FinBERT isolates the effect of financial domain pre-training on downstream sentiment classification. RoBERTa also produces 768-dimensional [CLS] embeddings.

The six feature sets span the full spectrum from sparse symbolic representations to rich contextual embeddings, and were designed to expose the relative contribution of each representational paradigm when combined with the classifiers.

Feature set	Type	Dim	Train shape	Val shape
TF-IDF Unigram	Sparse, count-based	5,097	(7,627 × 5,097)	(1,908 × 5,097)
TF-IDF Bigram	Sparse, count-based	9,873	(7,627 × 9,873)	(1,908 × 9,873)
Word2Vec (corpus)	Dense, static	100	(7,627 × 100)	(1,908 × 100)
GloVe Twitter-100	Dense, static	100	(7,627 × 100)	(1,908 × 100)
FinBERT [CLS]	Dense, contextual	768	(7,627 × 768)	(1,908 × 768)
RoBERTa [CLS]	Dense, contextual	768	(7,627 × 768)	(1,908 × 768)

The feature extraction results confirmed several points worth recording. The effective TF-IDF vocabulary settled at **5,097 unigrams** and **9,873 unigram+bigram features**, well below the configured max_features ceilings of 20,000 and 30,000 respectively. This reflects the natural compactness of the corpus after preprocessing: stop-word removal, lemmatisation, and the min_df=2 frequency threshold collectively reduce surface-form variety to a modest size, consistent with a domain-specific short-text corpus centred on a narrow set of financial concepts. The corpus-trained Word2Vec model produced a vocabulary of **5,214 tokens**, almost identical to the TF-IDF unigram count, since both apply the same minimum frequency threshold, confirming that both representations operate over the same effective lexical space and making their downstream classification results directly comparable. All six transformer embedding extractions completed successfully, producing the expected 768-dimensional [CLS] vectors for FinBERT and RoBERTa across training, validation, and test sets.

5. Classification Models

Two families of classification models were implemented: traditional ML classifiers trained on top of the feature representations produced, and fine-tuned transformer models trained end-

to-end on the raw tweet text. Within the traditional ML family, four classifier types were evaluated, Logistic Regression (LR), Random Forest (RF), XGBoost (XGB), and Multilayer Perceptron (MLP), each combined with the six feature sets and tested with and without balanced class weights, yielding a broad experimental grid. Fine-tuned transformers represent the upper bound of the experimental pipeline, where the feature extractor and classifier are trained jointly.

All models were evaluated on the held-out validation set (1,908 examples) using macro-averaged F1-score as the primary metric, with accuracy, per-class precision, recall, and F1 reported for completeness. Class weights were computed from the training set distribution, yielding weights of approximately 2.20 (Bearish), 1.65 (Bullish), and 0.52 (Neutral).

5.1 Logistic Regression

Logistic Regression was used as the primary baseline classifier, chosen for its interpretability, computational efficiency, and well-understood behaviour under class imbalance. It was combined with all six feature sets and evaluated with and without balanced class weights. Regularisation strength was tuned via cross-validation.

The results reveal a clear and consistent performance hierarchy driven by feature representation quality rather than by the classifier itself. Corpus-trained Word2Vec performs worst (macro-F1 0.49 unweighted), as the ~7,600 training tweets are insufficient for Word2Vec to learn meaningful vector geometry where Bearish recall collapses to 0.15, with the model defaulting almost entirely to Neutral predictions. GloVe Twitter improves modestly to 0.56 by importing distributional knowledge from 2 billion external tweets, but mean pooling still discards word order and context. TF-IDF recovers strongly to 0.70–0.72 despite being a sparse representation, because its vocabulary directly encodes the discriminative financial terms identified in exploration. Transformer embeddings (FinBERT, RoBERTa) reach 0.77, capturing contextual semantics that neither TF-IDF nor static embeddings can represent.

The effect of class weights is representation-dependent. For TF-IDF, balanced weights produce a meaningful gain: macro-F1 rises from 0.70 to 0.72 and Bearish recall improves from ~0.50 to 0.65 at negligible accuracy cost. For transformer embeddings, weights shift the precision–recall trade-off without improving macro-F1 (0.77 → 0.76), suggesting the representations are already rich enough that the classifier can locate minority class boundaries without rebalancing. For Word2Vec, class weights actively amplify noise, as the underlying embeddings lack sufficient discriminative signal to benefit from upweighting.

5.2 Random Forest

Random Forest was evaluated on the two strongest feature sets from the LR experiments, FinBERT and RoBERTa CLS embeddings, as tree-based ensembles are better suited to dense continuous features than to sparse high-dimensional spaces.

RF + FinBERT achieves macro-F1 0.74–0.75, slightly below LR + FinBERT (0.77), suggesting that the linear decision boundaries of Logistic Regression are in fact well-suited to the geometry of transformer embedding space. The more striking finding is the performance of RF + RoBERTa: without class weights it reaches only macro-F1 0.59, with Bearish recall of 0.20, among the lowest observed across all experiments. This indicates that Random Forest struggles to partition the RoBERTa embedding space effectively under class imbalance, likely due to the majority-class dominance in individual tree splits. Class weights partially recover performance (0.63), but the gap relative to LR + RoBERTa (0.77) remains substantial, making RF the weakest classifier overall on these representations.

5.3 XGBoost

XGBoost was evaluated on FinBERT and RoBERTa embeddings. It outperforms Random Forest consistently, achieving macro-F1 0.77–0.78 with FinBERT and 0.75–0.76 with RoBERTa. The gradient boosting framework's sequential error correction is more effective than RF's parallel ensembling at recovering minority class samples in dense embedding space.

XGB + FinBERT with class weights (macro-F1 0.78) matches the best LR result on the same features, while slightly improving Bullish F1 (0.75 vs 0.74). Notably, XGB shows more stable behaviour across the weighted/unweighted conditions than either LR or RF, with macro-F1 varying by at most 0.01 which suggests that its internal boosting mechanism is less sensitive to the class imbalance than the other methods.

5.4 Multilayer Perceptron

The MLP was implemented in PyTorch with cross-entropy loss, using a class-weighted loss function for the balanced variant. It was evaluated on FinBERT and RoBERTa embeddings and achieves the strongest results among all traditional ML classifiers.

MLP + RoBERTa (unweighted) reaches macro-F1 0.79 and accuracy 85.12%, the highest validation performance in the traditional ML family. MLP + FinBERT matches at macro-F1 0.79 with slightly lower accuracy (84.01%). The MLP's ability to learn non-linear transformations of the embedding space provides an advantage over the linear LR boundary, particularly for Bullish classification where F1 reaches 0.77, the best among all traditional classifiers on that class.

For the MLP, class weights do not improve macro-F1 and in some configurations slightly reduce it (0.79 → 0.77 for FinBERT), mirroring the pattern observed with LR on transformer features: when the representation is sufficiently expressive, loss reweighting trades macro-F1 for per-class recall balance without net gain.

5.5 Fine-tuned Transformer Models

Four transformer models were fine-tuned end-to-end for sequence classification, with a classification head trained jointly with the encoder (or decoder) weights. All models used balanced class weights via cross-entropy loss, and the best checkpoint was selected by highest validation macro-F1 across training epochs. The light preprocessing pipeline was used for all transformer inputs.

FinBERT (financial domain encoder, macro-F1 0.82, accuracy 86.32%) establishes the fine-tuning baseline. Compared to LR on frozen FinBERT embeddings (0.77), end-to-end fine-tuning yields a 5-point macro-F1 gain, confirming that adapting the encoder weights to the classification objective captures task-specific patterns that frozen [CLS] embeddings cannot. Bearish recall reaches 0.81, a substantial improvement over any frozen-embedding approach.

GPT-2 (general-domain decoder, macro-F1 0.83, accuracy 87.11%) represents the extra credit decoder-based classification experiment (**Extra Classification Model**). Despite being architecturally designed for autoregressive generation rather than bidirectional encoding, GPT-2 fine-tuned for classification achieves competitive performance, surpassing FinBERT by 1 point in macro-F1. This demonstrates that decoder models, when fine-tuned with a classification head on the final token representation, can be effectively repurposed for discriminative tasks.

Twitter-RoBERTa (Twitter-domain encoder, macro-F1 0.84, accuracy 87.89%) improves further by combining RoBERTa's robust pre-training procedure with Twitter-specific domain adaptation (**Extra Feature Engineering**). Its vocabulary and pre-training corpus are aligned with the register of the target data, making it the most domain-appropriate encoder among the tested models.

RoBERTa-base (general-domain encoder, macro-F1 0.87, accuracy 89.78%) achieves the best overall result across all models and all experimental conditions. All three per-class F1 scores reach their highest values here: Bearish 0.83, Bullish 0.86, Neutral 0.93. The gap between RoBERTa-base and Twitter-RoBERTa (0.87 vs 0.84) is counterintuitive, a general-domain model outperforming a Twitter-adapted one, and it suggests that the broader and more diverse pre-training corpus of RoBERTa-base produces more transferable representations than narrow Twitter-domain adaptation for this financial sentiment task.

Three conclusions emerge from the full picture. First, end-to-end fine-tuning dominates: all four fine-tuned transformers outperform all traditional ML classifiers by at least 3 macro-F1 points, confirming that joint optimisation of representation and classification is more powerful than the frozen-embedding pipeline. Second, within traditional ML, the MLP is the strongest classifier, benefiting from non-linear transformation of the embedding space. Third, TF-IDF remains a remarkably competitive sparse baseline, LR + TF-IDF Unigram (0.72) outperforms LR on both Word2Vec variants by a wide margin, underscoring that for narrow-domain short-text classification, term frequency encodes more usable signal than distributional embeddings trained on general corpora.

6. Evaluation and Results

6.1 Evaluation Methodology

All models were evaluated on the held-out validation set (1,908 examples, stratified to preserve the original 15.1% / 20.2% / 64.7% class distribution). Four metrics were computed for each model: Accuracy, Precision, Recall, and F1-Score, reported both per class and as macro-averages.

Accuracy measures the proportion of correctly classified tweets across all classes. It is reported for completeness but is not the primary criterion, as the severe class imbalance (64.7% Neutral) means a degenerate classifier that always predicts Neutral would achieve ~65% accuracy without any genuine discriminative ability.

Precision (per class) measures the proportion of tweets predicted as a given class that truly belong to it. High precision indicates few false positives for that class.

Recall (per class) measures the proportion of true instances of a class that the model correctly identifies. Low recall on Bearish or Bullish indicates the model is missing real sentiment-bearing tweets, defaulting instead to the majority Neutral class.

Macro-averaged F1-Score is the primary evaluation metric throughout this project. It computes the harmonic mean of precision and recall independently for each class and then averages across classes without weighting by support. This treats Bearish, Bullish, and Neutral as equally important regardless of their frequency in the training data, which is the right approach for an imbalanced three-class problem where misclassifying a minority sentiment carries real consequences in a financial context.

6.2 Performance Across Model Families

The experimental results tell a coherent and interpretable story across all model families.

Traditional ML on static embeddings (Word2Vec, GloVe) represents the weakest tier, with macro-F1 ranging from 0.49 to 0.57. The core limitation is not the classifier but the representation: mean-pooled static embeddings discard word order and produce a single vector per tweet that conflates semantically opposite sentences if they share vocabulary. Bearish recall is particularly damaged, reaching as low as 0.15, because the Bearish class is both the smallest and the most lexically similar to Bullish since they share earnings-related vocabulary but with opposing polarity that mean pooling cannot preserve.

Traditional ML on TF-IDF recovers substantially, reaching macro-F1 0.72 with Logistic Regression. This result is instructive: despite being a bag-of-words method with no semantic awareness, TF-IDF captures the discriminative financial vocabulary identified in exploration, *beats*, *misses*, *raised*, *cut*, which directly as high-weight features. In a narrow-domain corpus where key sentiment differences are lexically encoded in a small number of terms, term frequency is a highly effective signal. Bigrams add marginal value over unigrams (0.69 vs 0.70 unweighted), suggesting that the most informative n-grams in this corpus are single tokens rather than multi-word phrases.

Traditional ML on transformer embeddings (frozen FinBERT and RoBERTa CLS vectors) occupies the middle tier, with macro-F1 ranging from 0.59 to 0.79. The MLP is the strongest classifier in this family (0.79), followed by XGBoost (0.78) and Logistic Regression (0.77). Random Forest is the weakest (0.59–0.75), particularly struggling with RoBERTa embeddings under class imbalance. The consistent advantage of the MLP over linear classifiers confirms that the transformer embedding space has non-linear structure that a single linear boundary cannot fully exploit.

Fine-tuned transformers represent the strongest tier, with macro-F1 ranging from 0.82 (FinBERT) to 0.87 (RoBERTa-base). The 5-point gain of fine-tuned FinBERT over frozen FinBERT + LR (0.82 vs 0.77) isolates the contribution of end-to-end optimisation: adapting the encoder weights to the classification objective allows the model to reshape its internal representations around the specific distinctions that matter for this task, rather than relying on general-purpose [CLS] vectors.

6.3 The Role of Class Weights

Across the full experimental grid, the benefit of balanced class weights is representation-dependent and follows a clear pattern. For weak representations (Word2Vec), weights amplify noise and hurt performance. For moderate representations (TF-IDF), weights provide a meaningful macro-F1 gain (+0.02) by improving Bearish recall at acceptable precision cost. For strong representations (transformer embeddings used as frozen features), weights shift the precision–recall trade-off without net macro-F1 improvement. For fine-tuned transformers, all models used weighted cross-entropy loss, and the consistently high Bearish recall (0.80–0.82) across all four suggests this was the correct choice since that unweighted fine-tuning would likely have suppressed minority class performance as we could observe in the frozen-embedding experiments.

The general principle that emerges is that class reweighting compensates for representation weakness but cannot substitute for it: a well-trained contextual encoder reduces the need for explicit rebalancing because it learns class-discriminative features that are robust to frequency differences.

6.4 Per-Class Analysis

Bearish (class 0) is consistently the most difficult class to classify across all models and feature sets. It has the smallest support (15.1%), the highest lexical overlap with Bullish (shared earnings vocabulary), and the most contextual dependency, the difference between *"earnings beat"* and *"earnings miss"* can hinge on a single word whose significance only contextual models can fully capture. Fine-tuned transformers are the only models to achieve Bearish F1 above 0.74, with RoBERTa-base reaching 0.83.

Bullish (class 1) shows intermediate difficulty. It benefits from distinctive vocabulary (*beats*, *raised*, *buy*) and the cashtag frequency signal identified in exploration, but shares enough surface forms with Neutral (announcement-style text about positive corporate events) to cause confusion. Bullish F1 peaks at 0.86 with fine-tuned RoBERTa-base.

Neutral (class 2) is the easiest class to classify across all models, as expected from its dominant frequency and its distinct newswire register. Even the weakest model (LR + W2V Own, unweighted) achieves Neutral F1 of 0.81. Fine-tuned RoBERTa-base reaches 0.93. The high Neutral recall in weak models (0.93 for LR + W2V Own) confirms the earlier observation that these models are defaulting to Neutral rather than genuinely discriminating between classes.

6.5 Final Model Selection

The fine-tuned RoBERTa-base model was selected as the final model for test set prediction, having achieved the highest validation macro-F1 (0.87) and accuracy (89.78%) across all experiments, with the most balanced per-class performance (Bearish F1 0.83, Bullish F1 0.86, Neutral F1 0.93). Its superiority over the domain-adapted Twitter-RoBERTa (0.84) and the financially specialised FinBERT (0.82) suggests that the breadth and diversity of RoBERTa-base's pre-training corpus produces more transferable representations for this task than narrower domain adaptation. The final pipeline consists of the light preprocessing step described previously followed by end-to-end fine-tuned RoBERTa-base with a classification head, trained with balanced cross-entropy loss.

7. Conclusion

This project developed and evaluated a suite of NLP models for three-class financial sentiment classification on Twitter data, spanning the full pipeline from exploratory analysis through preprocessing, feature engineering, and classification.

The exploratory analysis established that the corpus is severely imbalanced (64.7% Neutral) and that sentiment signal is concentrated in a small set of discriminative financial terms, with Neutral tweets representing a distinct newswire register rather than the absence of sentiment. These findings directly shaped all subsequent methodological choices: macro-averaged F1 as the primary metric, stratified splitting, cashtag preservation, and POS-aware lemmatisation.

The experimental results demonstrate a clear and consistent performance hierarchy. Static embedding methods (Word2Vec, GloVe) proved insufficient for this task, with mean pooling destroying the contextual cues on which financial sentiment depends. TF-IDF emerged as a strong sparse baseline (macro-F1 0.72), outperforming both static embedding approaches by a wide margin and confirming that discriminative term frequency is more informative than general-purpose distributional semantics for narrow-domain short-text classification. Frozen transformer embeddings paired with non-linear classifiers (MLP, XGBoost) achieved competitive results (macro-F1 up to 0.79), while end-to-end fine-tuned transformers consistently outperformed all other approaches.

The best model across all experiments was fine-tuned RoBERTa-base, achieving a validation macro-F1 of 0.87 and accuracy of 89.78%, with balanced per-class performance (Bearish F1 0.83, Bullish F1 0.86, Neutral F1 0.93). Its superiority over domain-adapted alternatives (Twitter-RoBERTa, FinBERT) suggests that the breadth of general-domain pre-training is more valuable than narrow domain adaptation for this task, provided end-to-end fine-tuning is applied.

Two broader lessons emerge from the experimental grid. First, representation quality is the dominant factor in performance, the choice of feature engineering method explains more variance in results than the choice of classifier. Second, class reweighting is most effective when the underlying representation has moderate discriminative power; with strong contextual representations, the model learns minority class boundaries without requiring explicit rebalancing.